

Part - D OBE Evaluation - CO2				[1 x 15 = 15]
Name	ID	Section		
CO2 Description: Analyze the fundamental algorithms used in computer graphics and be able to compare and evaluate them in particular situations.				
Problem Analysis (5)	Content Knowledge (5)	Completeness (5)	Total (15)	

The coordinates shown in the following table were generated after applying a line drawing algorithm which is time consuming due to rounding. Derive another algorithm which does not use floating point arithmetic.

X	Y	Xp	Yp
2.0	3	2	3
2.83	4	3	4
3.67	5	4	5
4.5	6	5	6
5.33	7	5	7
6.17	8	6	8
7.0	9	7	9

The another algorithm which does not use floating point arithmetic is Bresenham's Midpoint Line algorithm.

Derivation

$$y = mx + c; ax + by + c = 0$$

$$\Rightarrow y = \frac{dy}{dx}x + c$$

$$dyx - dx y + dx.c = 0$$

$$\therefore a = dy, b = -dx, c = dx.c$$

Decision parameter

$$M = x_p + 1, y_p + \frac{1}{2}$$

$$d = f(M) \\ = f(x_p + 1, y_p + \frac{1}{2})$$

East

$$M_{new} = x_p + 2, y_p + \frac{1}{2}$$

$$d_{new} = f(x_p + 2, y_p + \frac{1}{2})$$

$$\Rightarrow \Delta d = d_{new} - d_{old}$$

$$\Rightarrow \Delta d \times 2 = a \times 2$$

$$\Rightarrow dE = 2a$$

$$\therefore dE = 2dy$$

North East

$$M_{new} = x_p + 2, y_p + \frac{3}{2}$$

$$d_{new} = f(x_p + 2, y_p + \frac{3}{2})$$

$$\Rightarrow \Delta d = d_{new} - d_{old}$$

$$\Rightarrow \Delta d \times 2 = 2(a + b)$$

$$\Rightarrow dNE = 2(a + b)$$

$$\therefore dNE = 2(dy - dx)$$

Initial

$$d = f(x_0 + 1, y_0 + \frac{1}{2})$$

$$\Rightarrow d_i = ax_0 + a + by_0 + \frac{b}{2} + c$$

$$\Rightarrow d_i = a + \frac{b}{2} \quad [ax_0 + by_0 + c = 0]$$

$$\Rightarrow d_i \times 2 = (a + \frac{b}{2}) \times 2$$

$$\Rightarrow d_i = 2a + b$$

$$\therefore d_i = 2dy - dx$$

(c) Briefly explain the Halftone technique in computer graphics. [5]

In halftone technique it creates a continuous range of colors or shades using dots/pixels.

4. (a) Show the first five iterations that will be obtained to draw a line using Bresenham's midpoint line algorithm where A(7,8) is starting and B(15,14) is the ending point. [10]

$$\begin{aligned} d_i &= 2dy - dx \\ &= 12 - 8 \\ &= 4 > 0 \rightarrow NE(8,9) \\ d_1 &= d_{old} + dNE \\ &= 4 + 2(dy - dx) \\ &= 4 + 2(-2) \\ &= 0 \geq 0 \rightarrow NE(9,10) \\ d_2 &= d_{old} + dNE \\ &= 0 + 2(dy - dx) \\ &= 0 + 2(-2) \\ &= -4 < 0 \rightarrow E(10,10) \end{aligned}$$

$$\begin{aligned} d_3 &= d_{old} + 2dy \\ &= -4 + 2(6) \\ &= 8 > 0 \rightarrow NE(11,11) \\ d_4 &= d_{old} + dNE \\ &= d_{old} + 2(dy - dx) \\ &= 8 + 2(-2) \\ &= 4 > 0 \rightarrow NE(12,12) \end{aligned}$$

$$\begin{aligned} d_5 &= d_{old} + dNE \\ &= 4 + 2(dy - dx) \\ &= 4 + 2(-2) \\ &= 0 \geq 0 \rightarrow NE(13,13) \end{aligned}$$

$$\begin{aligned} [10] \\ dy &= 14 - 8 = 6 \\ dx &= 15 - 7 = 8 \\ dy - dx &= 6 - 8 \\ &= -2 \end{aligned}$$

(b) Define the following:

- Direct Coding: In direct coding there is ~~space~~ for every pixel to describe the colors. [5]
- Scan conversion: Last step of rasterisation. (Vector image to raster image)
- Additive Color Model: RGB (Translated light) $\rightarrow$ Use for display.
- Subtracting Color Model: CMY (Reflected light) $\rightarrow$ Use for print.
- Color Palette: Finite set of colors.

In vector image if the image is modified then will be no change in image size and shapes. Because it made by mathematical equation and geometrical curves instead of dots or pixels/dots.

Applications: Logo, poster design, any image created with above illustration.

We assume (b) Given the radius of a circle is 7 and the center of the circle is (2,3), using Bresenham's Midpoint Circle Algorithm find the co-ordinates of the circle. Show only the first 5 iterations. [10]

Initial: (0, 2) = (0, 7)

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$$dE = d_0 + 2x_p + 3$$

$$= -6 + 0 + 3$$

$$= -3 < 0 \Rightarrow E(2, 7)$$

$$dE = d_0 + 2x_p + 3$$

$$= -3 + 2 + 3$$

$$= 2 > 0 \Rightarrow SE(3, 6)$$

$$dSE = d_0 + 2(x_p - y_p) + 5$$

$$= 2 + 2(2 - 7) + 5$$

$$= -3 < 0 \Rightarrow E(4, 6)$$

$$dE = d_0 + 2x_p + 3$$

$$= -3 + 6 + 3$$

$$= 6 > 0 \Rightarrow SE(5, 5)$$

$$dSE = d_0 + 2(x_p - y_p) + 5$$

$$= 6 + 2(4 - 6) + 5$$

$$= 7 > 0 \Rightarrow SE(6, 4)$$

0-1

(0, 7)
(1, 7)
(2, 7)
(3, 6)
(4, 6)
(5, 5)

for center (2, 3)

(2, 10)
(3, 10)
(4, 10)
(5, 9)
(6, 9)
(7, 8)

3. (a) For the given equation of a circle $(x-2)^2 + (y+7)^2 = 121$

- Find the center of the circle - (2, -7)
- Find the radius of the circle - $\sqrt{121} = 11$
- Initial coordinate of the circle - (0, 11)

(b) Suppose you have 3 coordinates of a circle A(3,10), B(4,9) and C(5,9). Identify the decision that was made moving from A to B. Also show the derivation for this formula. [5]

A(3,10), B(4,9), here we can see that from A to B the difference between (x,y) is x++, y--, so we can say the here used South East.

South East

$$M_{new} = x_p + 2, y_p - \frac{3}{2}$$

$$d_{new} = f(x_p + 2, y_p - \frac{3}{2})$$

$$\Delta d = d_{new} - d_{old}$$

$$\Delta d \times h = -h + 2(x_p - y_p) + 5$$

$$\therefore dSE = -h + 2(x_p - y_p) + 5$$

Decision Parameter

$$M = x_p + 1, y_p - \frac{1}{2}$$

$$d = f(M)$$

$$= f(x_p + 1, y_p - \frac{1}{2})$$

Part - C

Answer any 3 out of the 4 questions

[3 x 15 = 45 Marks]

1. (a) Consider the equation, $(x-2)^2 + (y+1)^2 = 9$ [5]
- State the type of image that can be made using this equation.
 - Explain how the image in (i) is made and displayed on the digital screen.

(b) Answer the following: [5]

- Show mathematically how you can create different color palettes.
- Convert the RGB color code (255, 150, 50) to equivalent CMY color code

(c) Write down the function names that will be used for the following tasks in glut. [5]

- Keep rendering events run in a loop –
- Change the window size –
- Send rendered images in the form of information into display –
- Apply color in a shape using color codes ranging from values 0-255 –
- Render images in the form of information before getting displayed –

2. (a) Suppose you need to design an image ensuring that it will hold its original quality despite going through multiple modifications in size and shape. Identify which type of image will be better suited according to the requirement and explain why. Also write some applications of the image type. [5]

Image type: Vector Image.

Explanation

In vector image, picture is created by some geometrical curves and mathematical equation instead of lots of pixels.

True/False - Darken A in the OMR if sentence is 'True' and B if 'False'

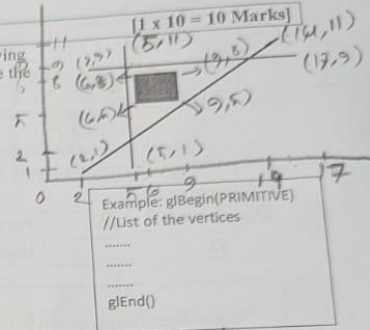
16. Raster files are significantly larger than vector files. **T**
17. Colors perceived in subtractive models are the result of translated light. **F**
18. Raster Image is better for designing a logo. **F**
19. Halftoning means converting halftone to raster image. **T**
20. Subtractive color models use CMYK color space. **F**

Part - B

Code Writing

Suppose you have to make the following design. Write the code showing only the vertex part along with the primitives that you will use to make the design.

Refer to the example given below to find the way you will answer-



// for the triangle border

glBegin(GL_LINES);

glVertex2f(2,9);

glVertex2f(17,9);

glEnd();

glBegin(GL_LINES);

glVertex2f(5,11);

glVertex2f(15,11);

glEnd();

glBegin(GL_LINES);

glVertex2f(2,1);

glVertex2f(14,11);

glEnd();

// for the rectangle

glBegin(GL_POLYGON);

glVertex2f(6,8);

glVertex2f(9,8);

glVertex2f(9,5);

glVertex2f(6,5);

glEnd();



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science & Technology
Department of Computer Science

CSC 4118: Computer Graphics (All Sections)

Mid-Term Examination

SET - A

Fall 2024-25

Total Marks: 100

Moderator: Aneem Al Ahsan Rupai

Time: 2 hours

Instructions:

- 1) There are 4 Parts [Part-A, Part-B, Part-C and Part-D]. Answer from all the Parts.
- 2) Answer Part-A (MCQ & True/False) in the OMR and other parts in the spaces provided in this question.
- 3) Part-D is for OBE evaluation. Answering this part is **MANDATORY**. MUST answer in the provided space inside the question paper.
- 4) Use of calculator is **ALLOWED** during examination.
- 5) Students may take **only** if required.

Name: [REDACTED]	Proctors Sign: [REDACTED]
ID: [REDACTED]	Permit No: [REDACTED] 7
Section: [REDACTED]	Department: CSE

Part - A MCQ [20 x 1.5 = 30 Marks]

1. The number of pixels stored in the frame buffer of a graphics system is known as:
A. Depth B. Resolution C. Image D. Pixel
2. During printing using 'Halftone' technique, the dots produced varies in:
A. Shape B. Size C. Space D. All of the above
3. More than 4 vertices in a primitive means a
A. Line B. Quad C. Triangle D. Polygon
4. What is the formula for South-East decision parameter in Bresenham's midpoint circle algorithm?
A. $d_0 + 2x_p + 3$ B. $1 - R$ C. $2(x_p - y_p) + 5$ D. $x^2 + y^2 - R^2$
5. Which one is a callback function?
A. glColorSub() B. glEnd() C. glutDisplayFunc() D. glFlush()
6. The complementary color of Red is:
A. Cyan B. Magenta C. Yellow D. Black
7. Which one is not a format of Raster graphics?
A. TIFF B. GIF C. PCX D. PICT
8. Which of the following is a property of incremental algorithm:
A. Slope B. Disjoint color C. Floating point D. Overlapping pixels
9. $(x+5)^2 + (y-7)^2 = 16$, considering the equation, which one of the following would be the initial Y coordinate according to Bresenham's midpoint circle algorithm?
A. 16 B. -5 C. 7 D. 4
10. What will be the North-East point of (4,8) according to In Bresenham's Midpoint Line algorithm?
A. (5,8) B. (5,9) C. (6,8) D. (5,7)
11. In coloring, the technique that uses a storage approach to identify the color codes is known as:
A. Direct Coding B. Lookup table C. Rasterization D. Halftone
12. Default value of leftmost co-ordinate in OpenGL with respect to gluOrtho2D() is:
A. 1 B. 0 C. -1 D. None
13. If $d \geq 0$, then which one is true for midpoint line algorithm?
A. $(\Delta d)_E = 2d_y$ B. $(\Delta d)_NE = 2(d_y - d_x)$ C. $d_{start} = 2d_y - d_x$ D. $(\Delta d)_{SE} = 2(d_y - d_x)$
14. The binary for color intensity values (128, 55, 167) is:
A. (10000000, 00101111, 01101011) B. (10000000, 00110111, 10100111)
C. (11000000, 00110111, 11010111) D. (01100000, 01000111, 10100111)
15. Which of the following statements is wrong:
A. Raster images depend on pixels B. Vector images have no intrinsic resolution
C. Raster images are less expensive than vector D. Vector images are malleable